

## Example 1:

$\text{king} - \text{man} + \text{woman} \approx \text{queen}$   
 $\text{\text{king}} - \text{\text{man}} + \text{\text{woman}} \approx \text{\text{queen}}$

- "king" and "queen" differ mainly by **gender dimension**.
- Subtracting "man" removes the *male component* from "king".
- Adding "woman" shifts it into the *female space*.
- Result  $\rightarrow$  a vector close to "queen".

## ♦ Example 2:

$\text{Paris} - \text{France} + \text{Italy} \approx \text{Rome}$   
 $\text{\text{Paris}} - \text{\text{France}} + \text{\text{Italy}} \approx \text{\text{Rome}}$

- "Paris" is the capital of "France".
- Removing "France" gives the **capital relationship vector**.
- Adding "Italy" shifts that relation to Italy.
- Result  $\rightarrow$  vector near "Rome".

## ♦ How It Looks (Conceptually in Vector Space)

Imagine embeddings as points:

- "king"  $\rightarrow [0.7, 0.2, 0.9, \dots]$
- "man"  $\rightarrow [0.6, 0.1, 0.7, \dots]$
- "woman"  $\rightarrow [0.5, 0.8, 0.4, \dots]$
- "queen"  $\rightarrow [0.6, 0.9, 0.6, \dots]$

Then:

$[0.7, 0.2, 0.9] - [0.6, 0.1, 0.7] + [0.5, 0.8, 0.4] \approx [0.6, 0.9, 0.6] \text{ (queen)}$   
 $[0.7, 0.2, 0.9] - [0.6, 0.1, 0.7] + [0.5, 0.8, 0.4] \approx [0.6, 0.9, 0.6]$   
 $(\text{\text{queen}})[0.7, 0.2, 0.9] - [0.6, 0.1, 0.7] + [0.5, 0.8, 0.4] \approx [0.6, 0.9, 0.6] \text{ (queen)}$

## Example 1

Sentence:

☞ *“I love my cat”*

### Step 1: Build Vocabulary

Unique words = {I, love, my, cat}

Vocabulary size = 4

### Step 2: Assign Index

- $I \rightarrow 0$
- $\text{love} \rightarrow 1$
- $\text{my} \rightarrow 2$
- $\text{cat} \rightarrow 3$

### Step 3: One-Hot Vectors

#### Word One-Hot Vector

I      [1, 0, 0, 0]

love [0, 1, 0, 0]

my    [0, 0, 1, 0]

cat    [0, 0, 0, 1]

## Example 2

Sentence:

☞ *“dog barks loud”*

### Step 1: Vocabulary

{dog, barks, loud}  $\rightarrow$  size = 3

### Step 2: Index

- $\text{dog} \rightarrow 0$
- $\text{barks} \rightarrow 1$
- $\text{loud} \rightarrow 2$

### Step 3: One-Hot Vectors

#### Word One-Hot Vector

dog    [1, 0, 0]

barks [0, 1, 0]

loud  $[0, 0, 1]$

### Example 3

Sentence:

↪ “*apple and mango*”

#### Step 1: Vocabulary

{apple, and, mango}  $\rightarrow$  size = 3

#### Step 2: Index

- apple  $\rightarrow 0$
- and  $\rightarrow 1$
- mango  $\rightarrow 2$

#### Step 3: One-Hot Vectors

##### Word One-Hot Vector

apple  $[1, 0, 0]$

and  $[0, 1, 0]$

mango  $[0, 0, 1]$